AI-BASED ALERT PRIORITIZATION FOR NETWORK INTRUSION DETECTION USING MACHINE LEARNING

CSCI 5742 – CYBER SECURITY PROGRAMMING AND ANALYTICS
SPRING 2026

Presented By:
Joseph Tesoriero
Tejal Jadhav

What is an Intrusion Detection System?

PROBLEM STATEMENT

Intrusion Detection Systems and Alert Volume:

- Organizations use Intrusion Detection Systems (IDS) and SIEM platforms to monitor network activity.
- These systems analyze network traffic and generate alerts when suspicious behavior is detected.

However, this creates a major challenges, Such As:

- A very large number of alerts are generated every day.
- Many alerts are false positives or low-priority events.
- This leads to alert fatigue for security analysts.
- As a result, important threats may be delayed or overlooked.

LIMITATIONS OF CURRENT IDS APPROACHES

Detection vs Prioritization:

- Many machine learning–based IDS systems focus on classification:
    - Label traffic as benign or malicious
    - Evaluate performance using metrics such as precision, recall, and F1-score
- However, classification confidence does not always reflect real-world impact.

Example:

- High probability attack on a low-value system
- Slightly lower probability attack on a critical server

The second alert may be more important to investigate first.

Problem Addressed in This Project:

There is currently no clear method to prioritize alerts based on operational risk.

This project proposes a system that:

- Uses a machine learning detection model
- Computes a risk score for each alert
- Combines:
    - prediction confidence
    - attack severity
    - asset importance
- Ranks alerts to highlight the most critical threats first.

MOTIVATION AND SIGNIFICANCE

Alert Overload in Cybersecurity Operations:

- Modern organizations generate a very high number of security alerts daily
- Many alerts are false positives or low-risk events
- This leads to alert fatigue for security analysts

Impact on Security Operations:

- Analysts must review thousands of alerts
- Important threats may be missed or delayed
- Security response becomes inefficient

Research Gap:

- Most IDS research focuses on improving detection accuracy
- Evaluations use metrics such as:
    - Precision
    - Recall
    - F1-score
- However, these metrics do not address how alerts should be prioritized

Motivation of This Project:

Develop a structured alert prioritization framework that combines:

- model prediction confidence
- attack severity
- asset importance

To rank alerts based on operational risk.

CORE COMPONENTS

1. Data Processing

- Clean and preprocess CICIDS2017 dataset
- Select features and split data for training/testing

2. Detection Module

- ML models (Random Forest, XGBoost)
- Classify traffic as benign or malicious
- Output malicious probability

3. Risk Scoring

- Combine:
    - model confidence
    - attack severity
    - asset importance

Generate final risk score

4. Alert Ranking

- Rank alerts by risk score
- Display highest-risk alerts first

METHODOLOGY

Dataset:

- CIC-IDS2017 dataset (network flow traffic)
- Labeled normal traffic and multiple cyber attack types
- Only CSV feature data used for experiments

Data Processing Pipeline:

1. Data preprocessing and cleaning
2. Feature normalization
3. Handling missing values (NaNs)
4. Addressing class imbalance between normal and malicious traffic
5. Model training and hyperparameter tuning
6. Risk score computation for alert prioritization

Machine Learning Models:

- Logistic Regression (baseline model)
- Random Forest
- XGBoost
- Support Vector Machine (SVM)

Goals:

- Detect malicious network traffic
- Rank alerts so analysts focus on the most critical threats first

EVALUATION

Classification Performance Metrics:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

Focus:

Reduce false positives, which are common in IDS systems.

Operational Performance Metrics:

1. Alert Ranking Effectiveness
   - Precision@K
   - Recall@K
   - Measures how many true attacks appear in the top K prioritized alerts
2. Detection Latency:
   - Measure time required for the model to generate a risk score for an alert
   - Used to estimate feasibility for real-time intrusion detection environments.

BASELINES

- FIFO Processing: alerts handled in chronological order
- Threshold-Based Detection: alerts flagged when probability exceeds a fixed threshold
- Logistic Regression: baseline machine learning model
- Probability-Only Ranking: alerts ranked by prediction probability without contextual risk weighting

RISK SCORE CALCULATION

RiskScore = P(Attack) * (Severity + AssetValue)

Where:

- P(Attack): Probability that traffic is malicious.
- Severity: Estimated severity of attack type (mapped from attack type label in CIC-IDS2017 dataset.
- Asset Value: Assigned based on simulated asset importance (inferred from destination IP ranges).

Goal:

- Rank alerts so that high risk attacks on critical systems appear first

Evaluation:

Compare:

- Probability only ranking
- Threshold
- FIFO
- Risk based ranking

Using Precision@K and Recall@K

CONCLUSION

Intrusion detection systems often generate a large number of alerts, many of which are false positives or low-priority events. This makes it difficult for security analysts to quickly identify the most critical threats. In this project, we propose a machine learning–based alert prioritization framework that combines detection probability with contextual factors such as attack severity and asset importance. By computing a risk score and ranking alerts accordingly, the system aims to reduce alert overload and help analysts focus on the most important security threats.